Technical Report

pylimma: a Python implementation of limma for differential expression analysis

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Summary

We present pylimma, a Python implementation of limma's linear-model workflow for differential expression analysis, covering voom precision weights, weighted linear model fits, empirical-Bayes moderated t- and F-statistics with the B-statistic, tests relative to a fold-change threshold (treat), contrasts and multiple testing. pylimma follows limma's algorithms step by step and is versioned against a specific limma release. On eight TCGA tumour–normal comparisons (4,972 samples) and 60 small-sample analyses, the limma-voom pipeline in pylimma returns the same differentially expressed genes as limma 3.68.5 with both the moderated t-test and treat, P-values within a relative difference of 1.6×10^{-4} (median 2.1×10^{-10}) and identical gene-set enrichment results, while running 1.2–1.3 times faster than limma on one thread. pylimma is the limma backend of OmicVerse and is released under LGPL-3.0-or-later at `https://github.com/omicverse/py-limma` and on PyPI as python-limma.

Keywords limma, voom, differential expression, RNA-seq, empirical Bayes, Python, reproducibility

Introduction

Differential expression (DE) analysis is the most common use of RNA-seq data[1], and limma[2] is one of the most widely used methods for it. limma fits a linear model to the expression of each gene and moderates the gene-wise variances towards a common or trended prior by empirical Bayes[3], which stabilises inference when the number of samples is small. With voom[4], which estimates the mean-variance relationship of log-counts and turns it into precision weights, the same linear-model machinery applies to RNA-seq counts, and limma-voom has become a standard pipeline for bulk RNA-seq[5].

Omics analysis is increasingly done in Python. Single-cell and spatial workflows are built on scanpy and the scverse ecosystem[6,7], and OmicVerse[8] brings bulk, single-cell and spatial analyses together in one Python framework. Bulk and pseudobulk DE is a routine step in these workflows, but limma is available only in R.

Python users currently reach limma through rpy2[9]. This requires a parallel R installation, converts data between the two languages at every call and exposes limma's options only through the binding layer. A native implementation removes all three costs, as PyDESeq2[10] has shown for DESeq2[11] and pyedger[12] for edgeR[13]. We set the same target for pylimma: an analysis in Python should return the same genes, the same P-values and the same pathways as the same analysis in R.

Here we present pylimma, a Python implementation of limma's linear-model workflow that is versioned against a specific limma release and reproduces its results to numerical precision. We describe its implementation and compare the limma-voom pipeline with limma 3.68.5 on eight TCGA cohorts and 60 small-sample analyses.

Implementation

Statistical model

limma fits, for each gene, a linear model of the expression values on the design matrix, by weighted least squares when observation weights are given. The residual variances of all genes are assumed to follow a scaled inverse-chi-squared prior, whose scale is constant or depends on average expression. `squeezeVar` estimates its degrees of freedom and scale from the data, and each gene's posterior variance combines its own residual variance with the prior[3]. Coefficients and contrasts are then tested with moderated t-statistics that use the posterior variances and the augmented degrees of freedom, with moderated F-statistics for several coefficients jointly, and with the B-statistic, the log posterior odds that a gene is differentially expressed[3]. `treat` tests coefficients against a fold-change threshold instead of zero[14]. For RNA-seq counts, voom computes log-counts per million from normalised library sizes, fits a lowess curve[15] to the square-root residual standard deviation against average log-count, and turns the

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fitted trend into a precision weight for every observation[4].

Features and code structure

pylimma 0.1.1 implements this workflow as in limma 3.68 (Table 1): `voom`, with adaptive lowess span and optional blocking; `lmFit`, with observation weights, duplicate spots and correlated samples through `duplicateCorrelation`; `contrasts.fit`; `eBayes`, with abundance trend and robust hyper-parameter estimation[16]; `treat`; `topTable`, `decideTests` and `removeBatchEffect`. As in R, `lmFit(v, design)` takes the voom object directly, and every function is exported under a Python name (`lm_fit`) and its R name (`lmFit`), so a limma script translates line by line. pylimma builds on NumPy[17], SciPy[18] and statsmodels[19], and returns results as pandas[20] data frames with limma's column names.

Numerical parity with limma

pylimma ports limma's algorithms instead of re-deriving the statistics. These include voom's choice of lowess span and its linear interpolation of the fitted trend, the per-gene weighted least-squares fits, `fitFDist` and the trigamma inversion that estimate the prior, the robust variant of `squeezeVar`[16] and the limit on the total degrees of freedom. For the B-statistic, `tmixture` estimates the prior variance of non-zero coefficients from the largest t-statistics, bounded by `stdev.coef.lim`. As in limma, this bound enters only the B-statistic; the moderated t-statistic uses the unscaled standard deviations as fitted. Each step is tested against reference output of limma 3.68.5 (9 tests), including a design with a continuous covariate, and pylimma versions follow limma releases (0.1.x tracks limma 3.68.5).

Use in OmicVerse

pylimma is the limma backend of OmicVerse[8]: `ov.bulk.pyDEG` runs the limma-voom pipeline with `deg_analysis(..., method='limma')`, tests continuous traits with `continuous_deg` and time courses with `timecourse_deg`, all on TMM-normalised[21] library sizes from pyedger[12].

Comparison with limma on TCGA

Data and design

We compared primary tumours with solid-tissue normal samples in eight TCGA projects[22] (BRCA, COAD, KIRC, LIHC, LUAD, LUSC, PRAD and THCA), using STAR gene counts from the UCSC Xena GDC hub[23] with one aliquot per participant (421 to 1,208 samples per cohort, 4,972 in total). We also drew 20 random subsets each of 3, 5 and 10 normal samples versus as many tumours from TCGA-LUAD; the data are those of the pyedger report[12]. Both packages ran the limma-voom pipeline of the limma user's guide: edgeR's `filterByExpr` and TMM normalisation (pyedger in Python), `voom` with design ~group, `lmFit`, then `eBayes` for the moderated t-test and `treat` with a log2 fold-change threshold of 1. Between 22,730 and 27,801 genes were retained per cohort. Genes were called DE at FDR ≤ 0.05 and $|\log_2 \mathrm{FC}| \geq 1$ for the moderated t-test, and at FDR ≤ 0.05 for treat. We compared four criteria: DE calls, gene-set enrichment, the numerical agreement of the statistics and run time (Supplementary methods).

Differential expression and pathways

Across the eight cohorts limma's moderated t-test called 56,192 DE genes, and pylimma's calls were identical in every cohort (Fig. 1a); treat (32,277 genes) also gave identical calls, and no gene crossed the FDR threshold in one package but not the other. The $-\log_{10} P$ values of the two packages lie on the identity line (Fig. S1). Gene-set enrichment with fgsea[24] on the MSigDB Hallmark collection[25] gave identical calls for all 400 cohort-pathway pairs (261 significant), with normalised enrichment scores (NES) differing by at most 1.1×10^{-8}, far below fgsea's own seed-to-seed variation (Fig. 1b).

Numerical agreement and speed

The statistics agree to the precision of voom's lowess fit (Table 1, Fig. 1c). The empirical-Bayes prior degrees of freedom and prior variance agree to 1.9×10^{-10} and 1.4×10^{-9}, and voom's precision weights to 1.4×10^{-6}. Per gene, log-fold changes agree to 1.2×10^{-5} in relative terms.

Moderated t-test P-values agree within 1.6×10^{-4} (median 2.1×10^{-10}), the largest differences occurring for the genes with the largest t-statistics, and B-statistics within 1.6×10^{-4}. In the 60 small-sample analyses the largest relative P-value difference was 4.6×10^{-6}, and all DE calls were identical (Fig. 1e).

pylimma ran the full pipeline 1.2 to 1.3 times faster than limma, both on one thread (Fig. 1d). For TCGA-BRCA, the largest cohort (1,208 samples), limma took 37 s and pylimma 29 s.

Conclusion and future perspectives

pylimma brings limma's linear-model workflow to Python with results that match limma to numerical precision. On eight TCGA cohorts of 421 to 1,208 samples and on 60 small-sample analyses, its limma-voom pipeline returns the same DE genes and pathways as limma 3.68.5 with both the moderated t-test and treat, and it runs faster than limma. Because pylimma works on pandas data frames and is part of OmicVerse, together with pyedger for count preprocessing, limma analyses can run in the same Python session as the rest of a bulk, single-cell or spatial workflow.

Future work includes the parts of limma beyond the core linear-model workflow: `voomWithQualityWeights` and `arrayWeights`, the gene-set tests `camera`, `roast`, `fry` and `goana`, the moderated F-test table `topTableF`, the microarray preprocessing functions and the plotting functions. pylimma will continue to follow limma releases, with each version pinned to the limma release it reproduces.

Data and code availability

pylimma is released under LGPL-3.0-or-later on PyPI (`pip install python-limma`; the import name is `pylimma`) and at https://github.com/omicverse/py-limma; this report describes version 0.1.1. The benchmark (`report/scripts` in the repository, with the task builder of py-edgeR) reproduces every number and figure in this report. TCGA STAR counts were obtained from the UCSC Xena GDC hub (https://gdc-hub.s3.us-east-1.amazonaws.com) and MSigDB Hallmark 2020 gene sets from Enrichr, both accessed on 9 November 2026. Software: pylimma 0.1.1 and pyedger 0.2.3 (Python 3.10, NumPy 2.2.6, SciPy 1.15.2); limma 3.68.5, edgeR 4.10.5 and fgsea 1.36.2

Table 1. limma functions and their pylimma counterparts (R names are exported as aliases).

limma	pylimma	Largest difference from limma 3.68.5 on TCGA
voom	voom	weights 1.4×10^{-6}
lmFit	lm_fit	residual SD 6.5×10^{-7}
eBayes (prior)	ebayes	df.prior 1.9×10^{-10}, s2.prior 1.4×10^{-9}
eBayes (t-test)	ebayes	P-value 1.6×10^{-4}
eBayes (B-statistic)	ebayes	B 1.6×10^{-4} (absolute)
treat	treat	P-value 1.1×10^{-4}
topTable	top_table	same order (sorted by B)

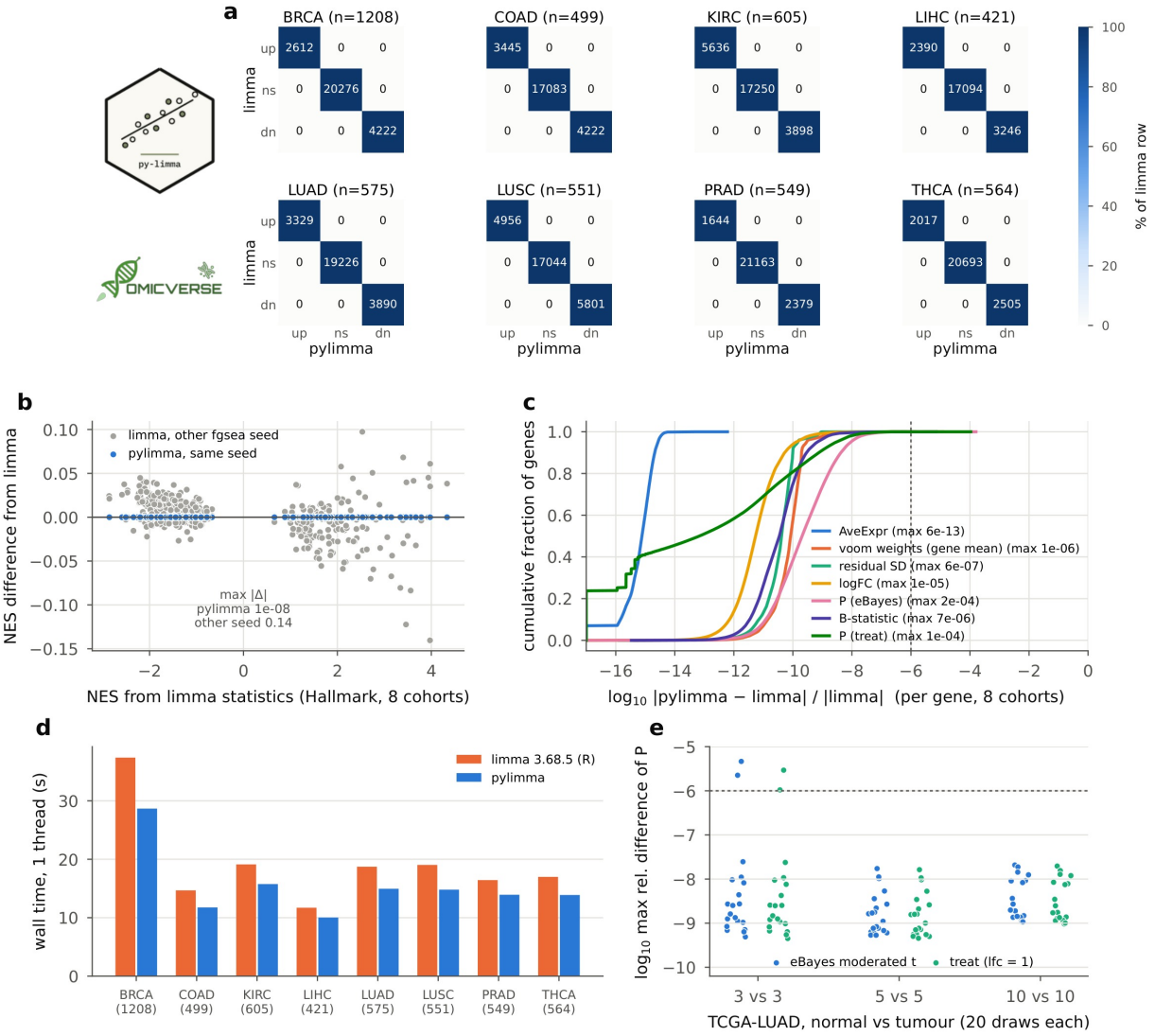


Figure 1. pylimma 0.1.1 versus limma 3.68.5 on TCGA tumour-normal comparisons.

(a) DE calls of the moderated t-test (FDR ≤ 0.05, $|\log_2 \mathrm{FC}| \geq 1$); numbers are genes, colour the percentage of the limma row. **(b)** Gene-set enrichment (fgsea, MSigDB Hallmark, 50 sets $\times$ 8 cohorts), genes ranked by the moderated t. Each point is one cohort-pathway pair: its NES from limma's statistics (x) and the difference from it (y) of the NES from pylimma's statistics with the same seed (blue) or from limma's statistics with another seed (grey). **(c)** Cumulative distribution of per-gene relative differences between pylimma and limma, pooled over the eight cohorts; values below 10^{-17} are drawn at 10^{-17}. **(d)** Wall time of the full pipeline, from normalisation to both tests, on one thread. **(e)** Largest relative P-value difference per random draw in the small-sample setting (TCGA-LUAD, 20 draws per size).

on R 4.5.3 (bioconda). Hardware: one node of the Stanford Sherlock cluster (Intel Xeon Platinum 8462Y+, 20 cores allocated).

Use of AI tools

Claude Opus 5.5 (Anthropic) was used to write the benchmark scripts and parts of the pylimma 0.1.1 code and tests, and to draft and edit the text. The authors reviewed all code, results and text and take full responsibility for the content of this publication.

Conflicts of interest

Z.Z. is affiliated with OmicVerse, Inc. The other authors declare no conflicts of interest.

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Supplementary information

Supplementary methods

Count data and small-sample analyses. The cohorts, sample selection and the 60 TCGA-LUAD subsets are those of the pyedger report[12]: integer counts were recovered from the $\log_2(\text{count}+1)$ values of the Xena hub, samples whose values were averages of replicate aliquots were removed, and each subset drew 3, 5 or 10 solid-tissue normal samples and as many primary tumours without replacement.

Pipeline. In R: `DGEList`, `filterByExpr`, `y[keep, , keep.lib.sizes = FALSE]`, `normLibSizes`, `voom(y, design)`, `fit <- lmFit(v, design)`, `eBayes(fit)` and `treat(fit, lfc = 1)`. In Python the count preprocessing used pyedger 0.2.3[12], which reproduces edgeR's filtering and TMM factors to 4.9×10^{-15}, followed by `voom` with the effective library sizes, `lmFit(v, design)`, `eBayes` and `treat` from pylimma. limma's defaults were used throughout, including voom's adaptive lowess span.

Gene-set enrichment. For each full cohort, genes were ranked by the moderated t-statistic, and fgsea was run on the 50 MSigDB Hallmark gene sets (minimum size 15, maximum size 500) once from limma's and once from pylimma's statistics with the same seed. Pathways were called enriched at adjusted $P \leq 0.05$. To show the scale of fgsea's Monte-Carlo variation, fgsea was also run on limma's statistics with a second seed, which changed 5 of the 400 calls and moved NES by up to 0.14 (Fig. 1b, grey).

Run time. Run times are wall-clock times of the full pipeline, from normalisation to both tests, with one run per cohort and package, one thread and BLAS pinned to one thread. limma and pylimma ran as one process per cohort in parallel on the same node, so that both were timed under the same load.

Notes for users coming from R

1. **Installation.** The PyPI distribution is `python-limma` and the import name is `pylimma`. An unrelated project publishes a package called `pylimma` on PyPI; install with `pip install python-limma`.
2. **Versions.** pylimma 0.1.x tracks limma 3.68.5, including its defaults (for example voom's adaptive lowess span).
3. **Indices are 0-based.** `topTable(fit, coef=1)` in Python is `coef = 2` in R; coefficient names are also accepted.
4. **Library sizes.** `voom` takes counts and the effective library sizes (library size $\times$ normalisation factor), for example from pyedger's `effective_lib_sizes`, where limma's `voom` reads them from a `DGEList`.

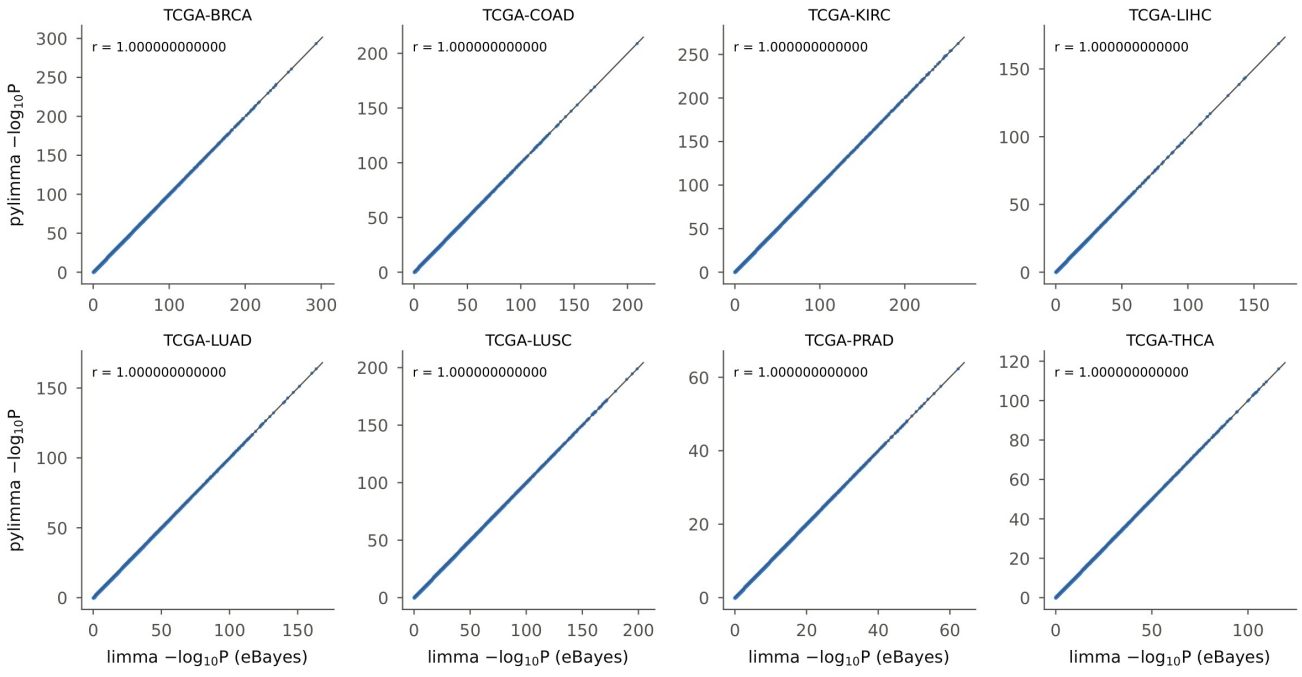


Figure S1. Moderated t-test P-values of pylimma against limma.

$-\log_{10} P$ of pylimma 0.1.1 (y) against limma 3.68.5 (x) for every retained gene of the eight TCGA cohorts; the line is the identity and r the Pearson correlation.